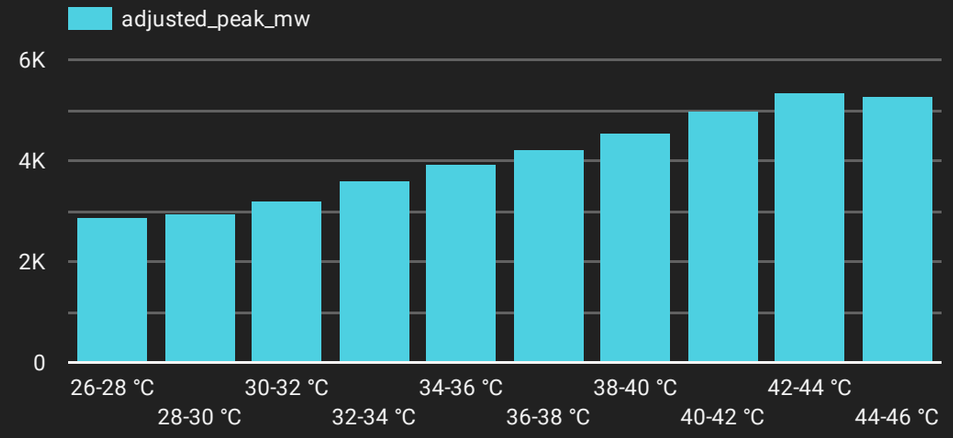
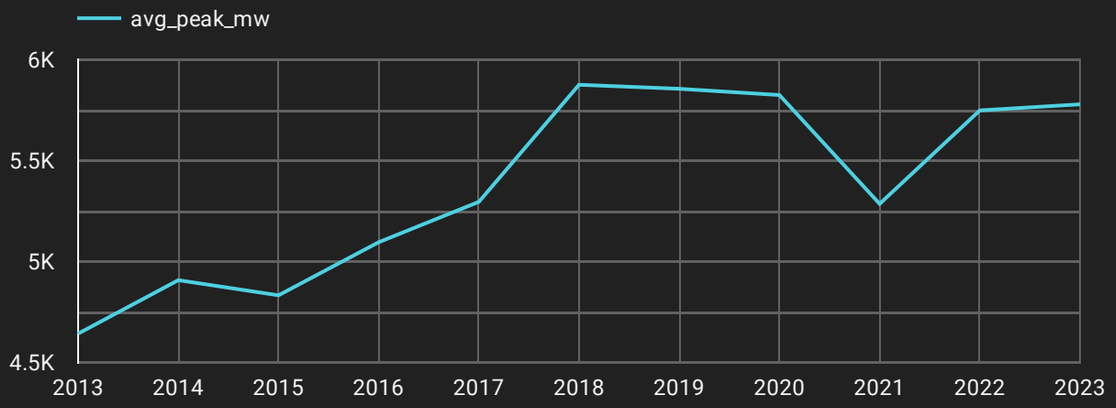
When will Delhi's power demand peak?

MW per extra °C of heat	MW growth per year	MW higher in Jul–Sep
171	115	1,203

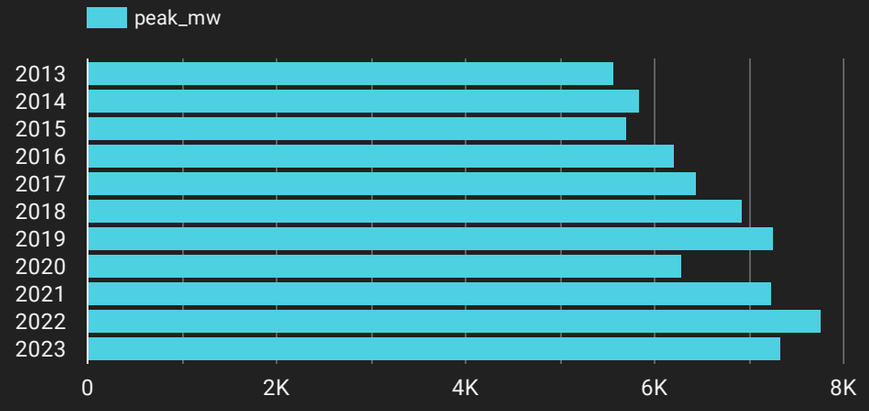
1. Demand rises steadily with heat from about 30 °C



2. Same heat (38–42 °C): demand rose to 2018, then flat



3. Yearly peak demand, 2013–2023



3b. Yearly peak dates (4 in June, 7 in July)

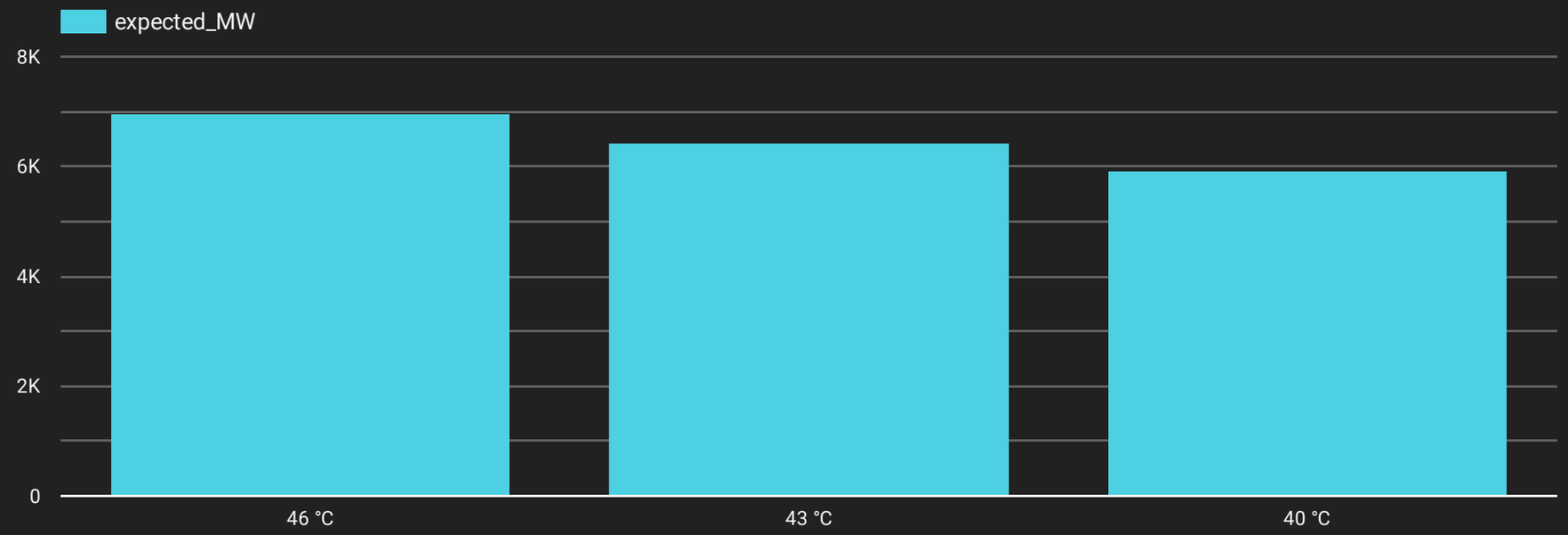
	year ▲	peak_date	month	peak_mw
1.	2013	Jun 6, 2013	June	5,567
2.	2014	Jul 15, 2014	July	5,852
3.	2015	Jun 10, 2015	June	5,710
4.	2016	Jul 1, 2016	July	6,208
5.	2017	Jun 6, 2017	June	6,441
6.	2018	Jul 9, 2018	July	6,924
7.	2019	Jul 2, 2019	July	7,254

Planning table: how much demand to plan for at 40 °C, 43 °C and 46 °C

	temp ▲	confidence	expected_MW	typical_day_low	typical_day_high
1.	40 °C	Medium	5,924	4,770	7,079
2.	43 °C	Medium	6,437	5,282	7,593
3.	46 °C	Low (no data at 46…	6,950	5,794	8,107

1 - 3 / 3

Expected peak demand at 40, 43 and 46 °C (MW)



Recommendation: *Plan for the upper end of each range, not the middle: about 7,100 MW at 40 °C, 7,600 MW at 43 °C and 8,100 MW at 46 °C (2023 level), plus a reserve for growth since. Prepare for the peak in June to July. 46 °C is a trend extension: no days at 46 °C in the data.*

Limits: *data ends April 2024. No humidity data. Temperature is from a weather-model grid, not the IMD station. $R^2 = 0.57$. 2020 lockdown removed. Sources: Grid-India via Hunt & Bloomfield (Zenodo 10.5281/zenodo.14983361); Open-Meteo*